KNIME WORKFLOW CREATION – Activity (Rose Mystica J- 192421410)

Problem Title: Student Academic Performance Analysis Using KNIME

An educational institution wants to analyze student academic performance using KNIME Analytics Platform. Create a complete KNIME workflow using an Excel dataset containing at least 30 student records with the following attributes:

Student_ID, Student_Name, Department, Gender, Attendance, Internal_Mark, Assignment_Mark, Model_Mark, Final_Mark, Study_Hours, Backlogs

Tasks

Create and execute a KNIME workflow to perform the following:

1. Data Import – 10 Marks
   Import the Excel dataset using Excel Reader and verify the data and data types.
2. Data Preprocessing – 10 Marks
   Identify and handle missing values using Missing Value node.
3. Column Selection – 5 Marks
   Select the relevant columns using Column Filter.
4. Attendance Filtering – 10 Marks
   Identify students having Attendance ≥ 75% using Rule-based Row Filter.
5. Total Mark Calculation – 10 Marks
   Create a new column Total_Mark using Math Formula/Column Expressions by calculating the total of Internal, Assignment, Model and Final marks.
6. Average Mark Calculation – 5 Marks
   Create Average_Mark by calculating the average of the four examination components.
7. Performance Classification – 10 Marks
   Using Rule Engine, classify students as:
   Excellent (≥80), Good (70–79), Average (60–69), Poor (<60).
8. At-Risk Student Identification – 10 Marks
   Using Rule Engine, classify a student as "At Risk" if Attendance <75 OR Average_Mark <50 OR Backlogs >2; otherwise classify as "Normal".
9. Department-wise Analysis – 10 Marks
   Using GroupBy, calculate for each department:
   Student Count, Average Attendance, Average Mark, Average Study Hours and Total Backlogs.
10. Sorting and Visualization – 10 Marks
    Use Sorter to arrange students according to Average_Mark in descending order. Create at least two visualizations showing department-wise average marks and performance-category distribution using suitable chart nodes.
11. Final Output – 10 Marks
    Display the final results using Table View and identify:
    - Top-performing students

- Poor-performing students
- At-risk students
- Department with highest average mark
- Department with highest total backlogs

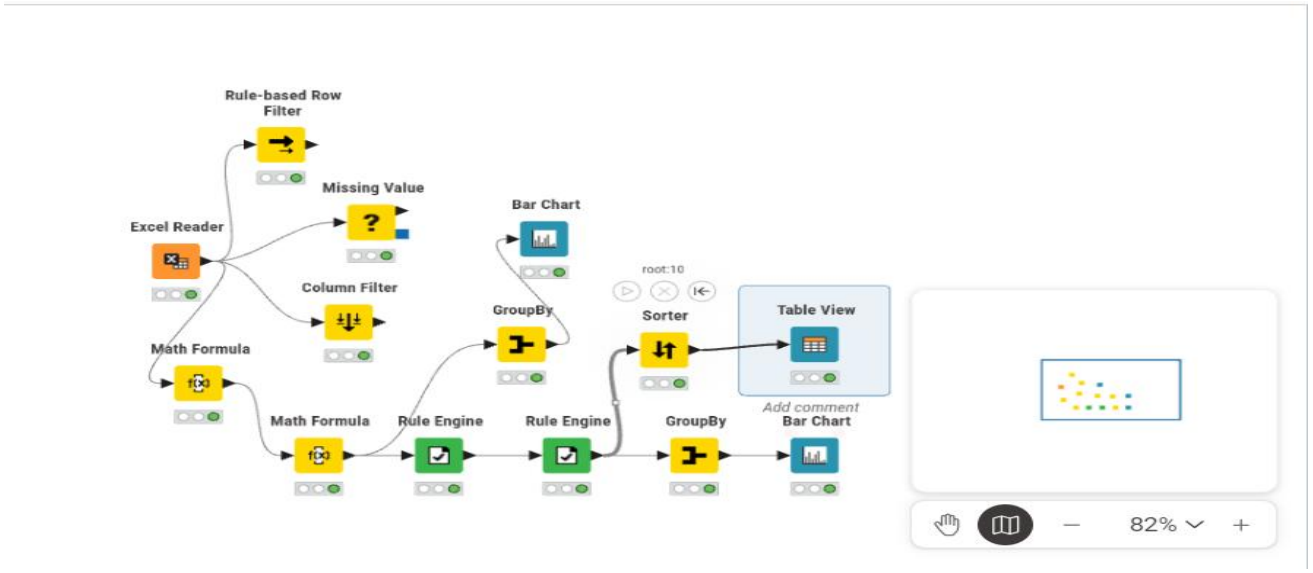
Minimum Required Nodes

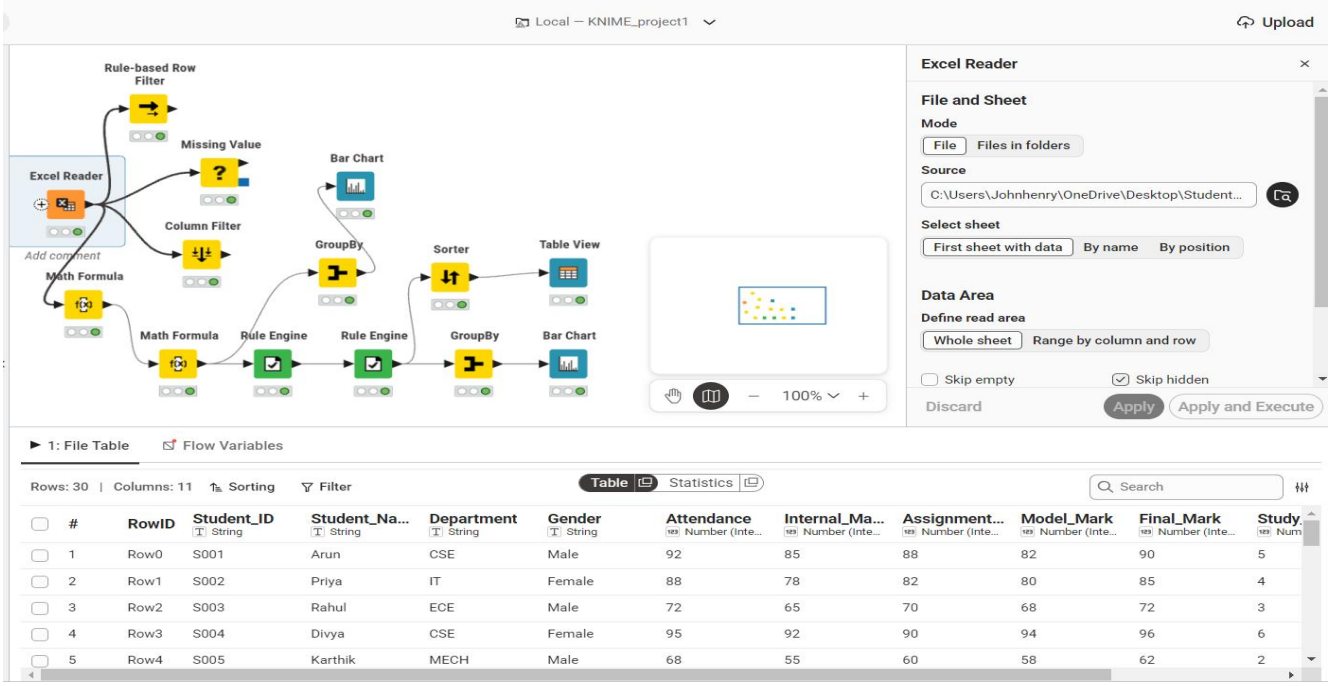
Excel Reader → Missing Value → Column Filter → Rule-based Row Filter → Math Formula/Column Expressions → Rule Engine → Sorter → GroupBy → Chart → Table View

Total: 100 Marks

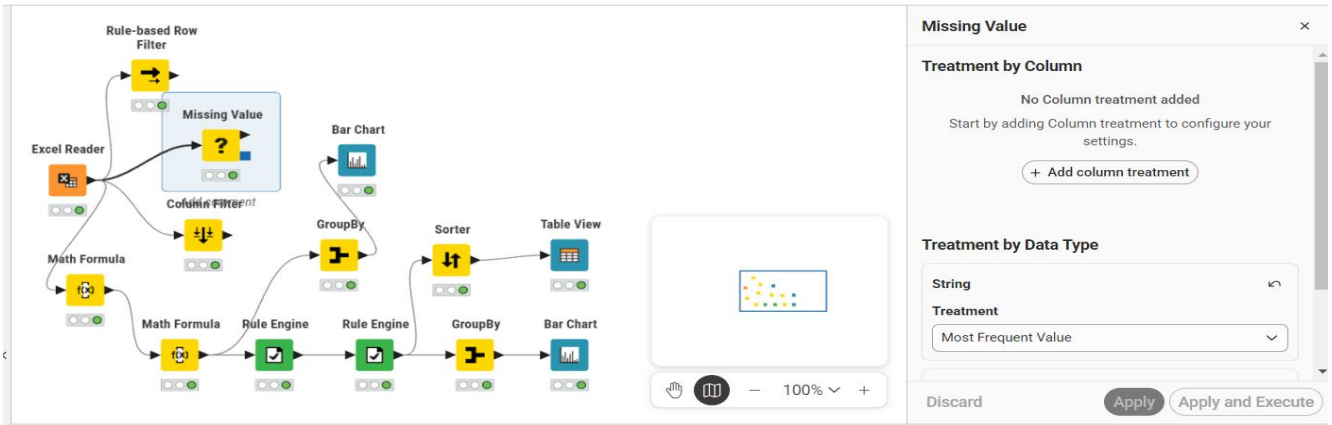
Instruction: Must independently configure, connect, execute and demonstrate the complete KNIME workflow and briefly interpret the obtained results.

OUTPUT:





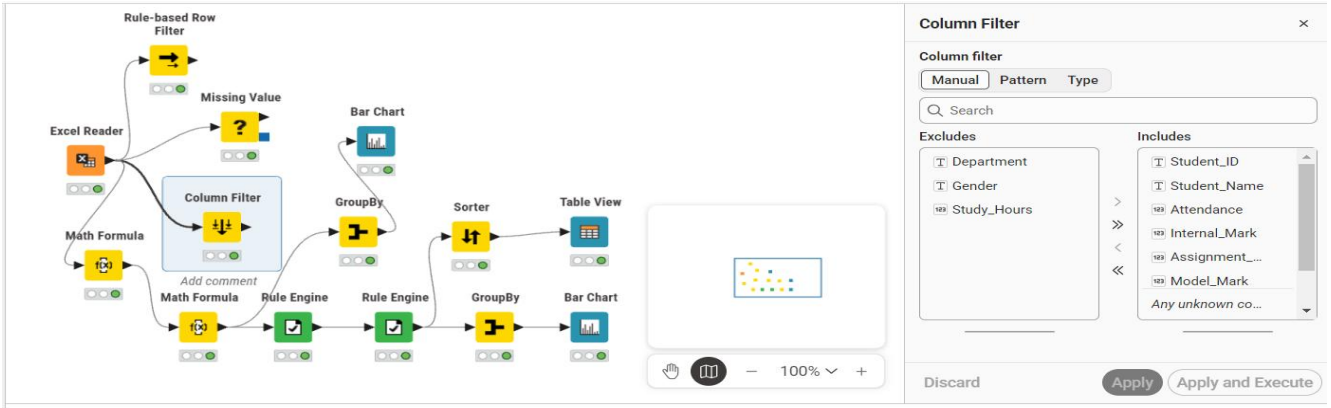
#	RowID	Student_ID	Student_Na...	Department	Gender	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Study
		String	String	String	String	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Num
1	Row0	S001	Arun	CSE	Male	92	85	88	82	90	5
2	Row1	S002	Priya	IT	Female	88	78	82	80	85	4
3	Row2	S003	Rahul	ECE	Male	72	65	70	68	72	3
4	Row3	S004	Divya	CSE	Female	95	92	90	94	96	6
5	Row4	S005	Karthik	MECH	Male	68	55	60	58	62	2



► 1: Output table ■ 2: PMML Transformations Flow Variables

Rows: 30 | Columns: 11 Sorting Filter Table Statistics Search

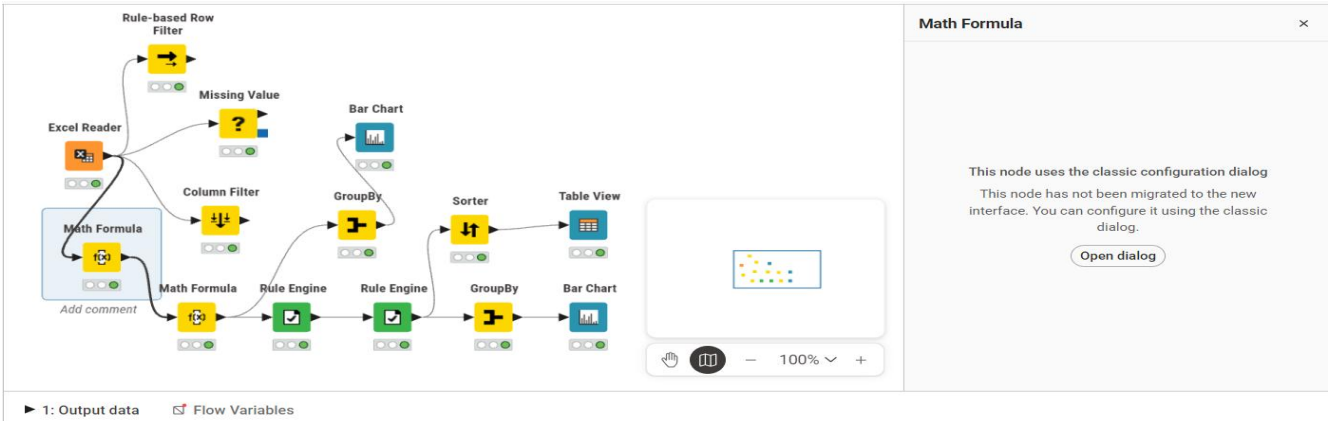
#	RowID	Student_ID String	Student_Na... String	Department String	Gender String	Attendance Number (Float)	Internal_Ma... Number (Float)	Assignment... Number (Float)	Model_Mark Number (Float)	Final_Mark Number (Float)	Study Num
1	Row0	S001	Arun	CSE	Male	92	85	88	82	90	5
2	Row1	S002	Priya	IT	Female	88	78	82	80	85	4
3	Row2	S003	Rahul	ECE	Male	72	65	70	68	72	3
4	Row3	S004	Divya	CSE	Female	95	92	90	94	96	6
5	Row4	S005	Karthik	MECH	Male	68	55	60	58	62	2



► 1: Filtered table Flow Variables

Rows: 30 | Columns: 8 Sorting Filter Table Statistics Search

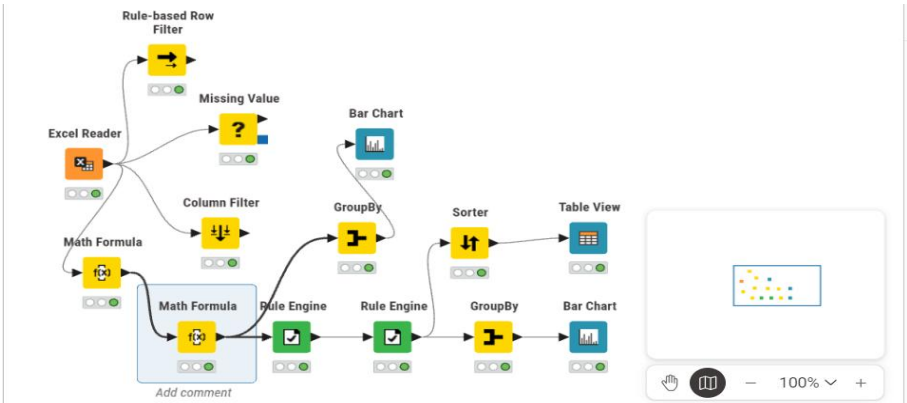
#	RowID	Student_ID String	Student_Name String	Attendance Number (Integer)	Internal_Mark Number (Integer)	Assignment_... Number (Integer)	Model_Mark Number (Integer)	Final_Mark Number (Integer)	Backlogs Number (Integer)
1	Row0	S001	Arun	92	85	88	82	90	0
2	Row1	S002	Priya	88	78	82	80	85	1
3	Row2	S003	Rahul	72	65	70	68	72	2
4	Row3	S004	Divya	95	92	90	94	96	0
5	Row4	S005	Karthik	68	55	60	58	62	3



► 1: Output data Flow Variables

Rows: 30 | Columns: 12 Sorting Filter Table Statistics Search

#	RowID	Student_ID String	Student_Na... String	Department String	Gender String	Attendance Number (Inte...	Internal_Ma... Number (Inte...	Assignment... Number (Inte...	Model_Mark Number (Inte...	Final_Mark Number (Inte...	Study Num
1	Row0	S001	Arun	CSE	Male	92	85	88	82	90	5
2	Row1	S002	Priya	IT	Female	88	78	82	80	85	4
3	Row2	S003	Rahul	ECE	Male	72	65	70	68	72	3
4	Row3	S004	Divya	CSE	Female	95	92	90	94	96	6
5	Row4	S005	Karthik	MECH	Male	68	55	60	58	62	2



Math Formula

This node uses the classic configuration dialog

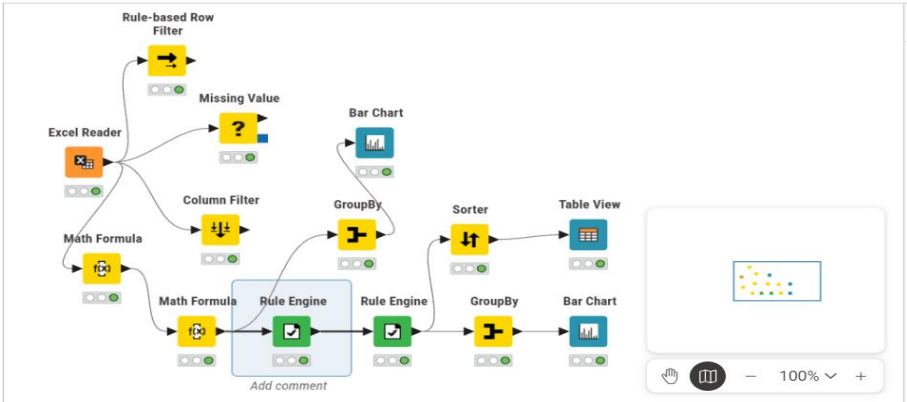
This node has not been migrated to the new interface. You can configure it using the classic dialog.

Open dialog

▶ 1: Output data | Flow Variables

Rows: 30 | Columns: 13 | Sorting | Filter | Table | Statistics | Search

artment	Gender	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Study_Hours	Backlogs	Total_Mark	Average_M...
	String	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Float)	Number (Float)
	Male	92	85	88	82	90	5	0	345	86.25
	Female	88	78	82	80	85	4	1	325	81.25
	Male	72	65	70	68	72	3	2	275	68.75
	Female	95	92	90	94	96	6	0	372	93
H	Male	68	55	60	58	62	2	3	235	58.75



Rule Engine

This node uses the classic configuration dialog

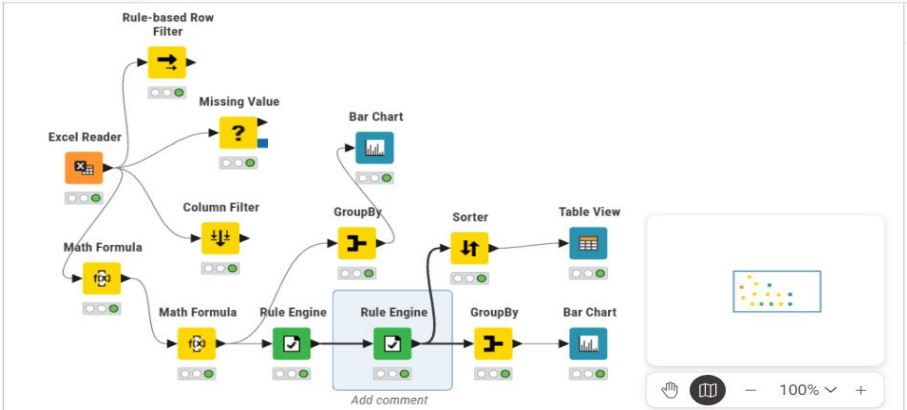
This node has not been migrated to the new interface. You can configure it using the classic dialog.

Open dialog

▶ 1: Classified values | Flow Variables

Rows: 30 | Columns: 14 | Sorting | Filter | Table | Statistics | Search

ider	Attendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Study_Hours	Backlogs	Total_Mark	Average_M...	Performance
tring	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Float)	Number (Float)	String
	92	85	88	82	90	5	0	345	86.25	Excellent
ale	88	78	82	80	85	4	1	325	81.25	Excellent
	72	65	70	68	72	3	2	275	68.75	Average
ale	95	92	90	94	96	6	0	372	93	Excellent
	68	55	60	58	62	2	3	235	58.75	Poor



Rule Engine

This node uses the classic configuration dialog

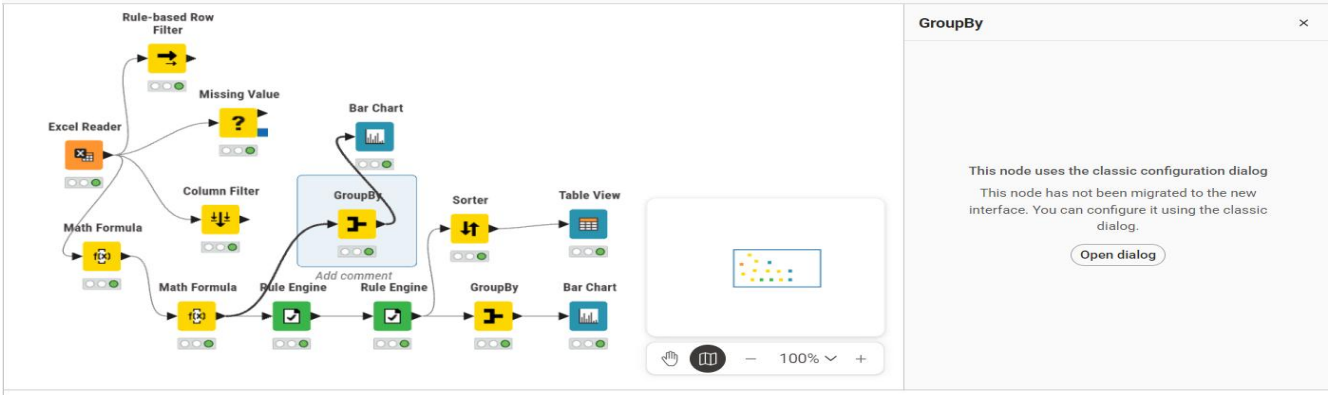
This node has not been migrated to the new interface. You can configure it using the classic dialog.

Open dialog

▶ 1: Classified values | Flow Variables

Rows: 30 | Columns: 15 | Sorting | Filter | Table | Statistics | Search

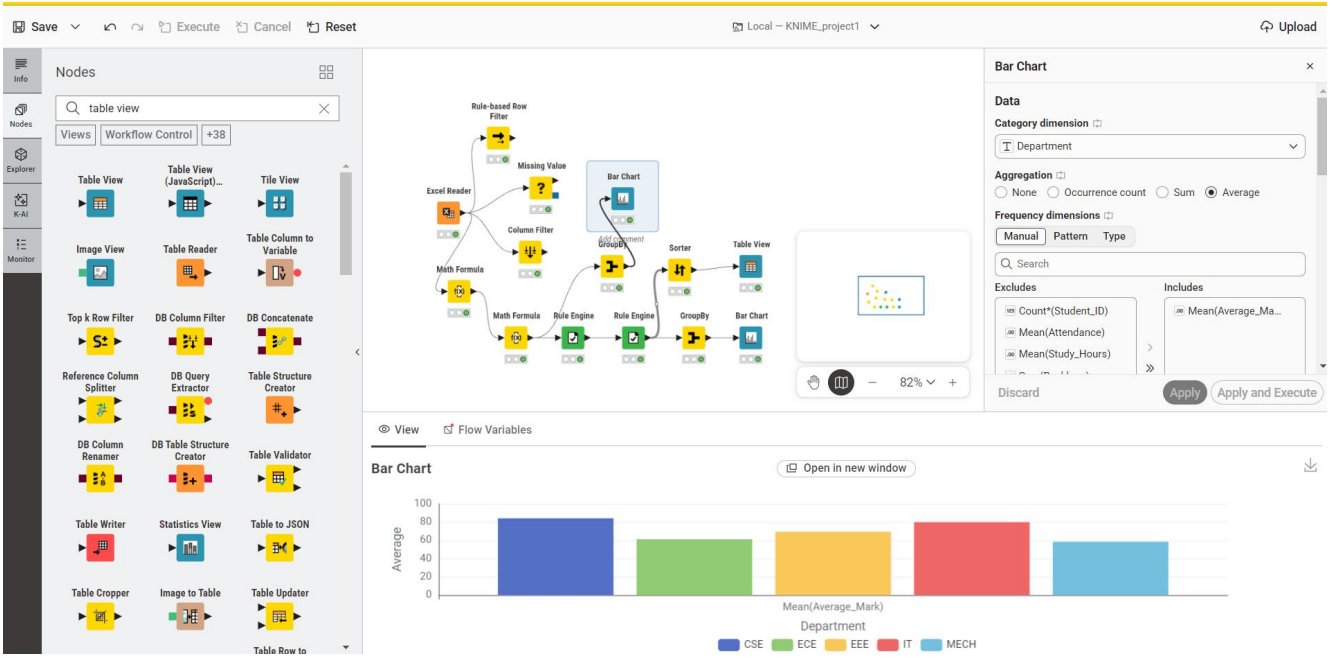
tendance	Internal_Ma...	Assignment...	Model_Mark	Final_Mark	Study_Hours	Backlogs	Total_Mark	Average_M...	Performance	Risk_Status
Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Float)	Number (Float)	String	String
	85	88	82	90	5	0	345	86.25	Excellent	Normal
	78	82	80	85	4	1	325	81.25	Excellent	Normal
	65	70	68	72	3	2	275	68.75	Average	At Risk
	92	90	94	96	6	0	372	93	Excellent	Normal
	55	60	58	62	2	3	235	58.75	Poor	At Risk

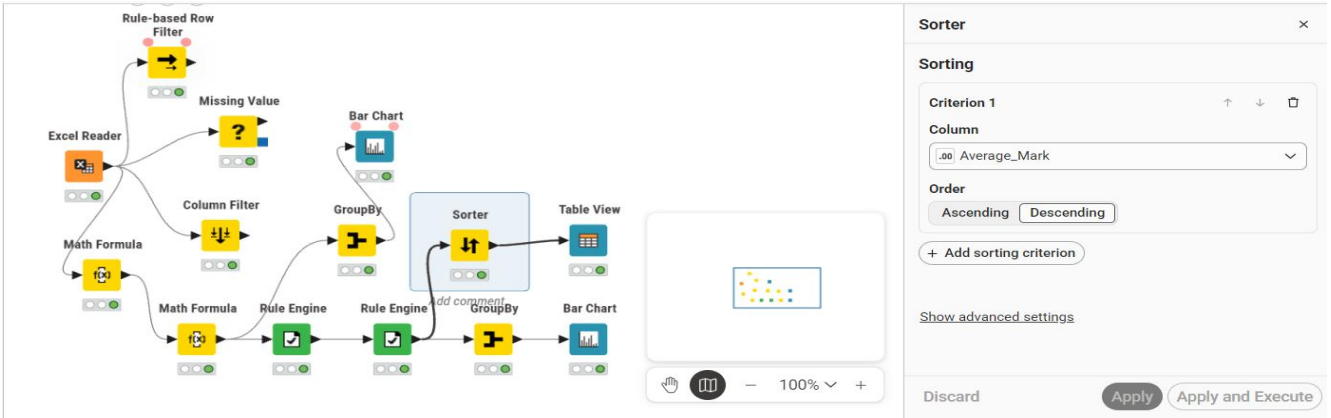


► 1: Group table Flow Variables

Rows: 5 | Columns: 6 Sorting Filter Table Statistics Search

#	RowID	Department (String)	Count*(Student_ID) (Number (Integer))	Mean(Attendance) (Number (Float))	Mean(Average_Mark) (Number (Float))	Mean(Study_Hours) (Number (Float))	Sum(Backlogs) (Number (Integer))
1	Row0	CSE	8	89.5	84.125	5.25	3
2	Row1	ECE	5	75	61.2	2.6	12
3	Row2	EEE	5	82.2	69.4	3.4	7
4	Row3	IT	7	88.286	79.929	4.857	4
5	Row4	MECH	5	72.8	58.45	2.4	12

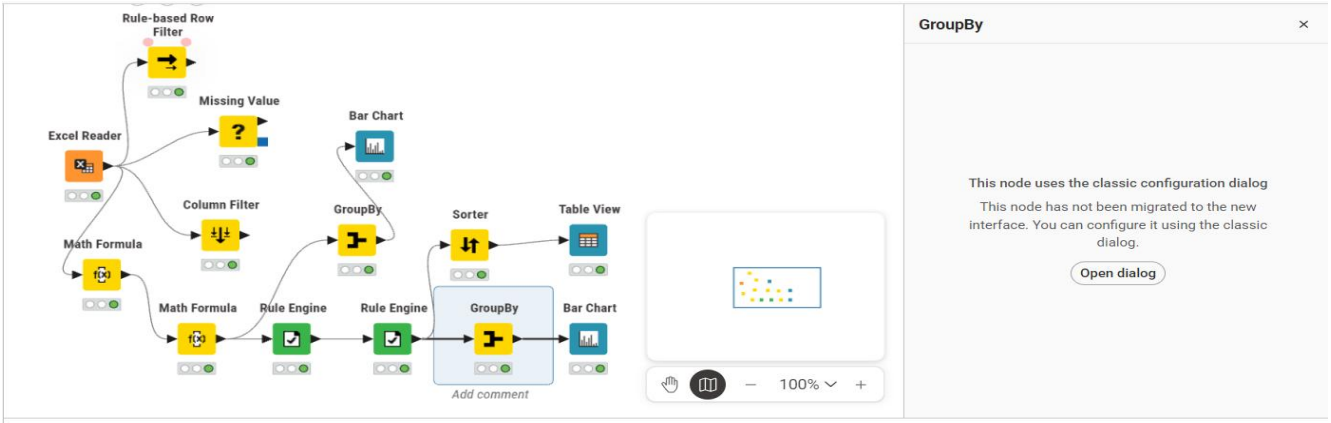




► 1: Sorted Table Flow Variables

Rows: 30 | Columns: 15 Sorting Filter Table Statistics Search

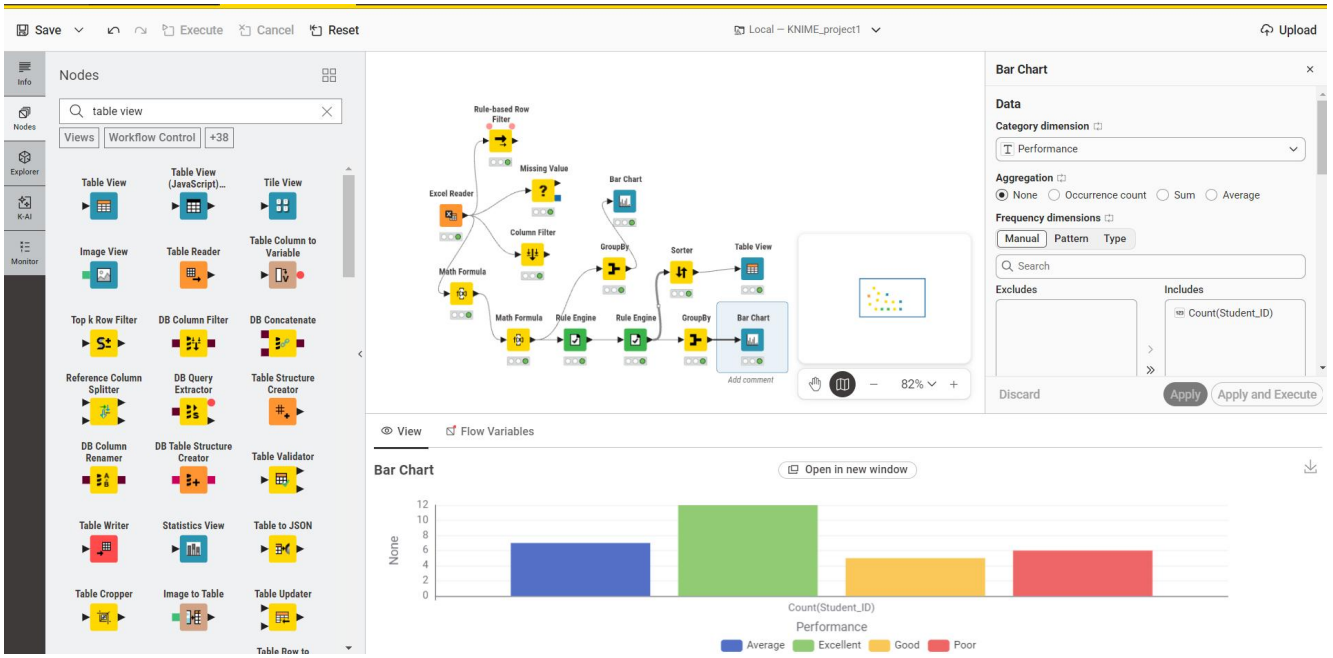
#	RowID	Student_ID (String)	Student_Na... (String)	Department (String)	Gender (String)	Attendance (Number (Inte...)	Internal_Ma... (Number (Inte...)	Assignment... (Number (Inte...)	Model_Mark (Number (Inte...)	Final_Mark (Number (Inte...)	Study... (Num
1	Row23	S024	Swetha	CSE	Female	98	94	96	95	98	8
2	Row9	S010	Meena	IT	Female	96	95	92	94	97	7
3	Row3	S004	Divya	CSE	Female	95	92	90	94	96	6
4	Row15	S016	Aishwarya	CSE	Female	94	90	93	91	95	6
5	Row7	S008	Anjali	CSE	Female	90	88	85	87	91	5

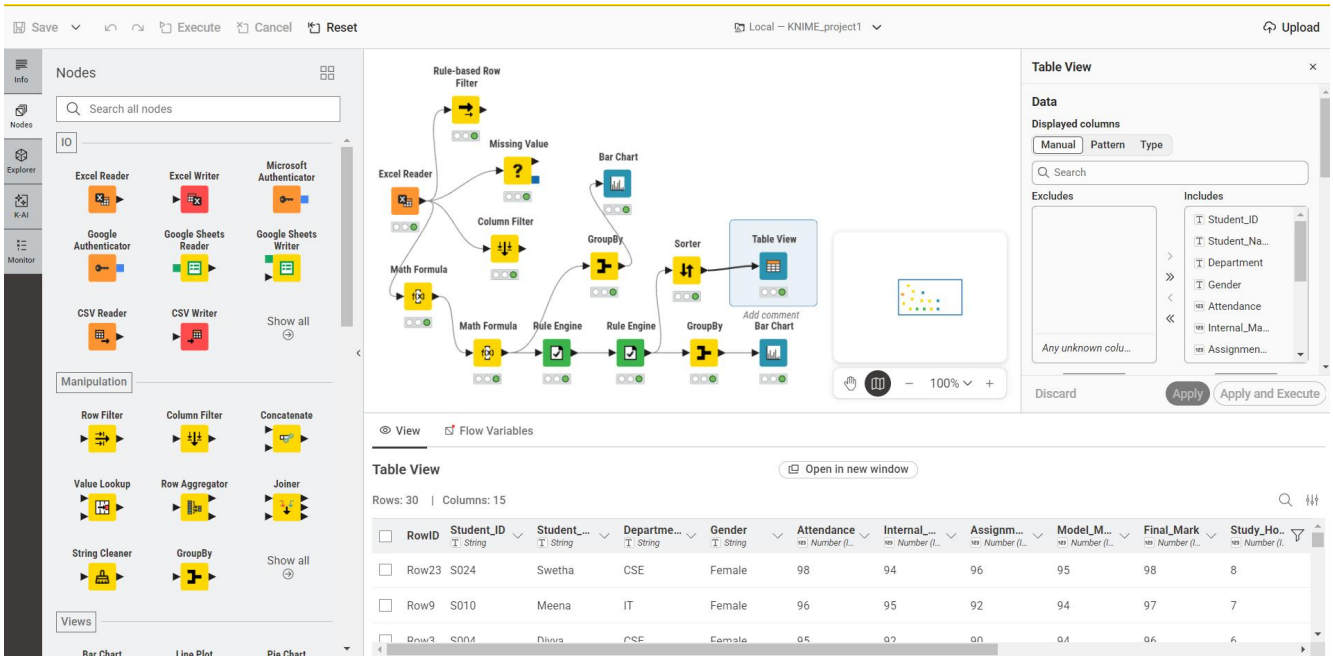


▶ 1: Group table | Flow Variables

Rows: 4 | Columns: 2 | Sorting | Filter | Table | Statistics | Search

#	RowID	Performance (String)	Count(Student_ID) (Number (Integer))
1	Row0	Average	7
2	Row1	Excellent	12
3	Row2	Good	5
4	Row3	Poor	6





RowID	Student_ID (String)	Student_... (String)	Departme... (String)	Gender (String)	Attendance (Number (I...)	Internal_... (Number (I...)	Assignm... (Number (I...)	Model_M... (Number (I...)	Final_Mark (Number (I...)	Study_Ho... (Number (I...)
Row23	S024	Swetha	CSE	Female	98	94	96	95	98	8
Row9	S010	Meena	IT	Female	96	95	92	94	97	7
Row3	S004	Divya	CSE	Female	95	92	90	94	96	6